

1.1 Operating System - Concept, Components, System, operations of OS, Program management, Resource management, Security and Protection.

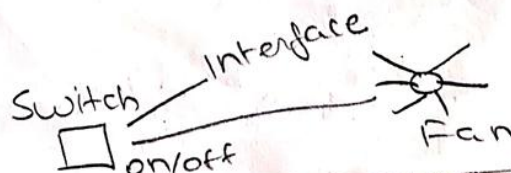
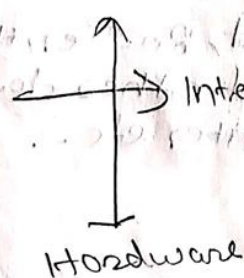
Operating System? (Action) (heart of CPU) Concept

It is a set of programs which controls all the computer resources and provide environment in which user can develop application programs.

An Operating System is a software act as interface between end-user and hardware.

It creates user-friendly environment.

If there is no operating system user can't interact with the hardware. If he wanted to do without O.S. he have to write program for each and every thing.



Operations of OS

Evolution/generation of operating system.

① First generation (1940-1956) - Vacuum tubes, Plug boards.

* These machines had Vacuum Tubes as a central component.

* Machines of the time were so primitive that programs were often entered one bit at a time on rows of mechanical switches (plug boards).

* But the vacuum tubes were having the problem of heating and they required the large space.

* There was no operating system in this generation.

* Programming languages were unknown (not even Assembly language)

* So all the programming was done in machine language. (Binary) 101010

* All the problems were simple numerical calculations.

Q] List any four functions of operating system.

(Any four functions - 1 mark each)

Ans: The major functions of an operating system are:

1. Process Management – A program in execution is called as Process. OS is responsible for creating, deleting, suspending and resuming the processes.

2. Memory Management – OS is responsible for allocation and deallocation of memory to the system programs as well as other programs and data.

3. File Management – An OS deals with the **storage, creation** and **deletion** of file on various storage devices. It also allows all files to be easily **changed and modified** through the use of text editors.

4. Device / Resource Management – Managing devices and resources and allowing the users to share the resources such as **CPU time, main memory, secondary storage and input output devices**.

5. Security and Protection – **Securing the system** against possible **unauthorized access** to data or any other entity by means of passwords and similar other techniques.

6. Job Accounting- Keeping track of **time and resources** used by **various jobs and users**.

7. Communications – Providing **interface for the user to communicate** with each other in the same or other system by sharing or transferring data.

1.2 Different types of Operating System: Batch Operating System, Multi-programmed, Time Shared Operating System, Multiprocessor System, Distributed System, Real Time System, Mobile OS (Android OS)

Batch Operating System

The users of Batch operating system do not interact with computer directly.

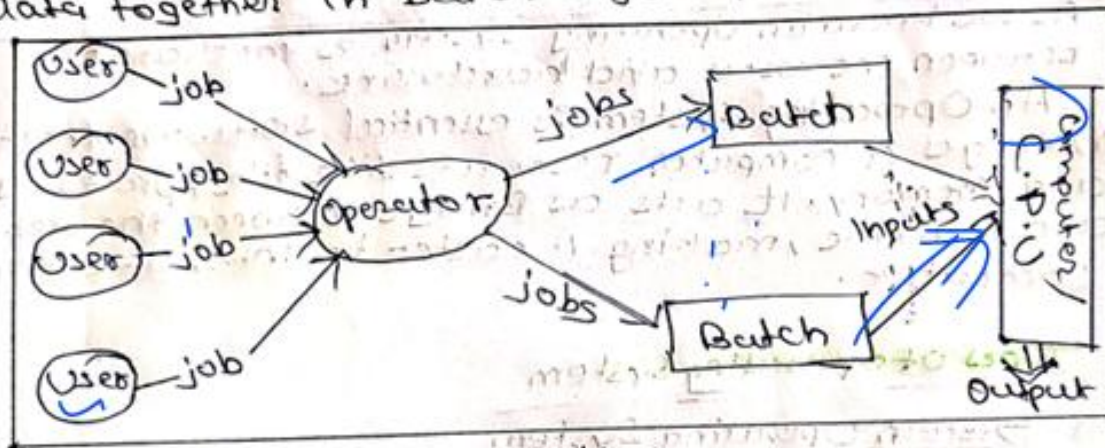
Each user prepares his job on offline device like punch cards and submit it to the Computer Operator (person).

The Batch operating system will work to submit similar kinds of jobs together.

Job: Job is collection of program, I/P data and Control Instruction.

In Batch operating system, a batch is a sequence of user jobs.

Here operating system collects program and data together in batch before processing starts.



It follows following activities

- ① Operating system defines a job which has predefined sequence of command, programs and data in single unit.
- ② Operating system keeps a no. of jobs in memory and executes them without any manual information.
- ③ Jobs are processed in order of submission i.e. First come first served.
- ④ After when jobs are executed the operator separates the similar kind of jobs in batch.
- ⑤ As an input Batches are given to Computer CPU it executes it in FIFO and process it one batch at a time & provide desired output.

Applications: Payroll System, Bank Statements etc.

Multiprogrammed Operating System

Multiprogramming operating system can be simply illustrated as more than one program present in main memory and one of them kept for in execution. This is basically used for better execution of resources.

Job 1

The operating system keeps several jobs in memory at a time.

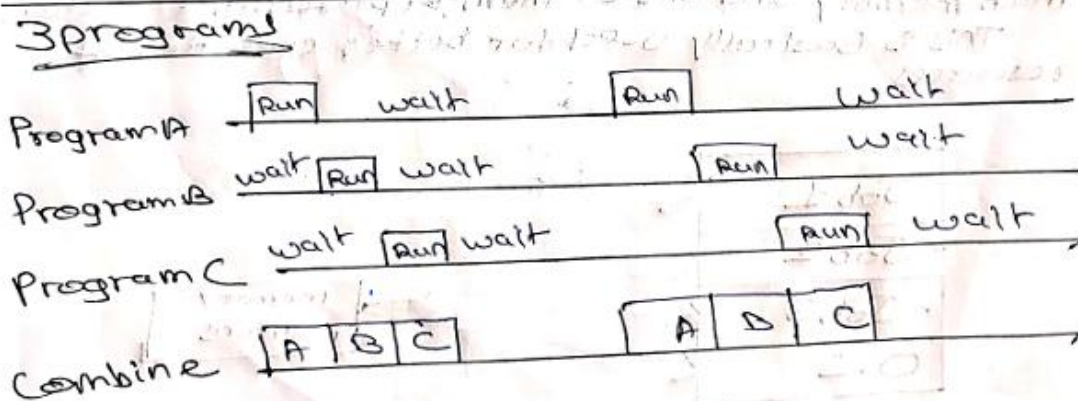
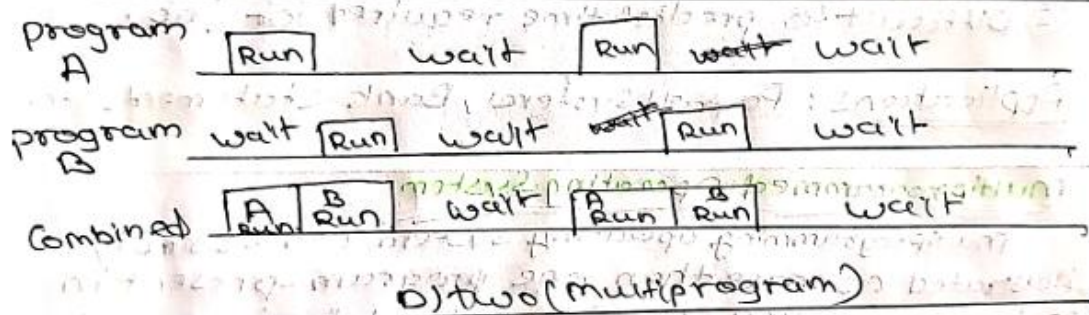
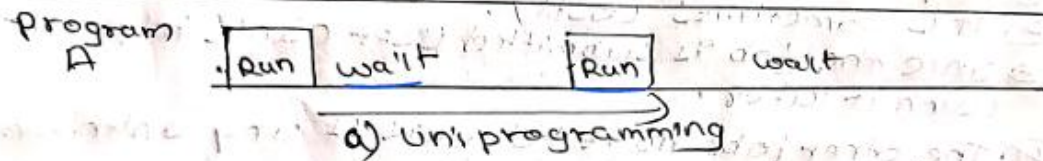
The operating systems keep several job in memory at a time.

O.s picks and begin to execute one of job in memory. Eventually the job may have to wait for I/O operation to complete.

In Non-Programming mode System: CPU will sit idle.

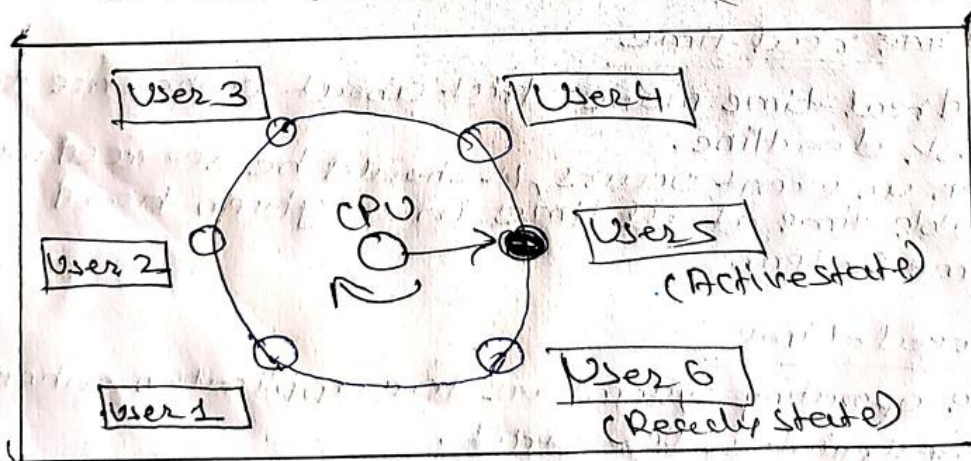
As long as there is always some job to execute, the CPU will never be idle.

Throughput of system will also increase.



Time sharing operating system

- Each task is given some time to execute so that all the tasks work smoothly.
- Each user gets the time of CPU as they use a single system.
- These system are also known as multitasking systems.
- The task can be from a single user or different users also.
- The time that each task gets to execute is called Quantum.
- After this time interval OS switches over to next task.



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Advantages

- Each task get equal opportunity.
- Fewer chances of duplication of software.
- CPU idle time can be reduced.
- Avoids duplication of software.

Disadvantages

- Reliability problem.
- Data communication problem.
- One must take care of the security and integrity of user programs and data.

9) Multiprocessor System

1) When more than one CPU are used in execution of the processes it is known as multiprocessor system.

2) Many CPU are assigned in one cabinet to work in one cabinet for closed communication.

3) Multiprocessor system are also known as Parallel Systems or tight coupled systems.

4) It is illustrated as multiprocessor which leads to better throughput rate.

5) Here CPU share Registers and cached memory and connected main memory.

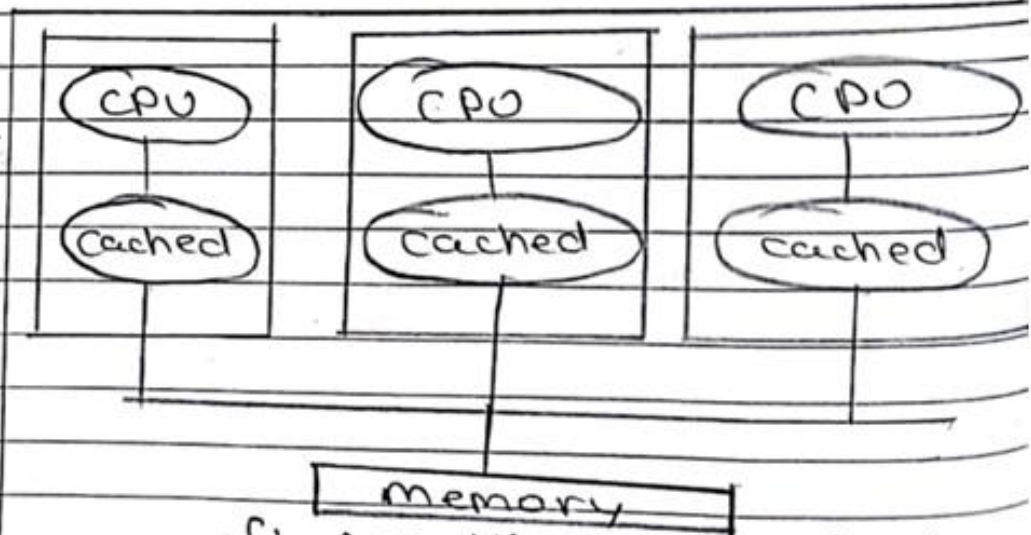


fig: multiprocessor system

There are two types of multiprocessor system.

- 1. Symmetric System
- 2. Asymmetric System.

1. Symmetric System

There is single multiple CPU which are treated equally.

No master-slave relationship.

2. Asymmetric System

There is one main CPU and other CPUs are followed or instructed by main. Follows master-slave relationship.

5. DISTRIBUTED SYSTEM A distributed system is a network that consists of autonomous computers that are connected using a distribution middleware. They help in sharing different resources and capabilities to provide users with a single coherent network. In this system, processor does not share memory or clock instead each processor has its own memory. Processors in a distributed system may vary in size and function. These processors are referred as sites, nodes, computers, and so on. The nodes communicate with each other through high speed buses or telephone lines by message passing. These systems are also referred as Loosely Coupled System.

These systems are also referred as **Loosely Coupled System**.

Distributed Operating System

```

graph TD
    subgraph Nodes
        direction TB
        N1["CPU  
Memory  
Disk"]
        N2["CPU  
Memory  
Disk"]
        N3["CPU  
Memory  
Disk"]
        N4["CPU  
Memory  
Disk"]
    end
    N1 <--> CN((Communication Network))
    N2 <--> CN
    N3 <--> CN
    N4 <--> CN
    style N1 fill:#fff,stroke:#000
    style N2 fill:#fff,stroke:#000
    style N3 fill:#fff,stroke:#000
    style N4 fill:#fff,stroke:#000
    style CN fill:#008000,color:#fff
  
```

Advantages:

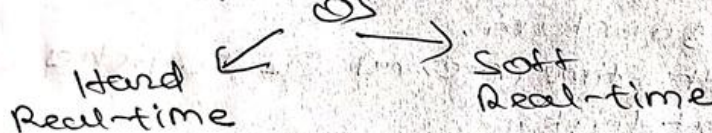
Real-time Operating Systems

These types of OS serve real time systems. The time interval required to process and respond to inputs are very small.

This time interval is called response time.

Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control, robots etc.

Types of Real-time OS



Hard real-time

Hard real-time means strict about adherence to each task deadline.

When an event occurs, it should be serviced within predictable time at all times in a given hard real-time system.

Soft real-time

These operating systems are for application when time constraint is less strict.

Applications

↳ Flight Control Systems

↳ Simulations.

<u>iOS</u>	<u>Android</u>
Apple is Developer	Google is Developer
Family - UNIX, OS X	OS Family - Linux
No, except of widgets except Notification center	Yes, there are widgets
Programmed in C, C++, C#	Programmed in C, C++, Java.
closed, with open source components	Open source
messaging - iMessage	messaging - Google Hangouts
Launched in 2007	Launched in 2008
Costly	Cheaper

Features	Time sharing	Real time O.S
<u>Definition</u>	A time sharing system allows several users to use a computer system from various location simultaneously.	A Real time O.S. completes a task within a specific time.
<u>Computer Resources</u>	User may share the resources between users.	Resources are allotted to a process for a fixed amount of time before reallocating to another process.
<u>Switching</u>	A switching method/function is available.	The switching method/function is not available.
<u>modification of program</u>	Any modification may be possible in program.	Any modification may not be possible in the program.
<u>Response</u>	The response is generated within seconds.	The user must get the response within specified time constraint.
<u>Process details</u>	It runs the multiple applications simultaneously.	It runs only single application simultaneously.
<u>Goal</u>	It focus is on responding quickly to a request.	It focuses on completing computer task before the deadline.

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<u>Multi programming / multiple</u> processes run concurrently at the same time on a single processor. It is based on the concept of context switching. It takes more time to execute the processes. It utilizes the single CPU for execution of processes. It takes more It speed is slow Easy to use	<u>Multi tasking</u> is when more than one task is executed at single time utilizing multiple CPUs. It is based on concept of time sharing. It takes less time to execute the tasks. It utilizes multiple CPU for execution of task. It Speed is fast, Difficult to use
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```

graph TD
    Job1 --> MP[memory partition]
    Job2 --> MP
    Job3 --> MP
    OS --> MP
        
```

```

graph LR
    Print --> OS[OS]
    Sort --> OS
    Email --> OS
    OS --> CPU[CPU]
        
```

High UV Now

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